Wind Tunnel

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Introduction to Engineering Design

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February 27, 2025

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Brainstorming

Water mill. Our first idea was to create a wheel that generates electricity from flowing water. To create a steady stream of water, we would 3D print a little circle, and using fans we could simulate a running river. The wheel would then turn while connected to a DC motor which would then transfer electricity through a wire to the object you need to power.

Rube Goldberg machine. Another idea was a Rube Goldberg machine which would simulate energy transfer through normal household objects.

Volumetric display. A volumetric display is a display that uses a rapidly rotating led matrix, usually at a few thousand rpm, to create an illusion of a 3D object displayed. The LEDs change multiple times with each rotation, in effect creating a 3D display that can be viewed at all angles. This idea was pushed aside due to the extreme complexity and difficulty in both the construction and the programming involved.

Flying car/drone. We wanted to create a drone that could also function as a car with wheels on the ground. We proposed to use cycloidal propellers since they are essentially cylinders that generate lift and would function as wheels, negating the need for two separate systems in drone mode or car mode. However, cycloidal propellers were too complex as the pitch of the propellers had to constantly change.

Wind tunnel. Our idea was to create a small tunnel which would use a fan to generate wind. A scale model would then be placed in the test section, and smoke would be released through tubes to visualize the airflow around the object. Furthermore, we would use force sensors or scales to measure the drag/lift or other aerodynamic properties of the test object.

Decision Matrix

We used a decision matrix to narrow down our initial ideas based on our criteria. We chose these criteria to determine the feasibility and complexity of our project as well as if it will fulfill the constraints. Some examples of our decisions are if the project had to use 3D printing, the complexity of the project, and if it can be completed by May 1st. Here is a link to our decision matrix.

Project Description

Our project is a scale wind tunnel designed for testing small models. The tunnel will consist of three main parts: the inlet, which is a wide opening for as much air to pass though; the test section, where the test model will be tested; and the diffuser, which will provide the air flow with a fan. The test section will consist of a stand to hold the model and scale/force sensors to measure the aerodynamic forces on the test object, allowing us to analyze the aerodynamics of the object. Also, smoke will be released from a smoke machine through tubes into the test section, allowing us to visualize the airflow around the object as well. The wind tunnel will mostly consist of 3D printed sections bolted together. The test section will include a clear plastic window for viewing, and the bottom of the test section will have openings for force measurement and serve to allow us to adjust the angle of attack while testing. The diffuser and inlet sections will also consist of plastic straws stacked into a grid to serve as air straighteners, reducing turbulence and ensuring the airflow will be as laminar as possible. Finally, an arduino will be used to control the fan and to display information such as airspeed, and a 12V power supply will provide power a to the arduino, fan, and smoke machine. This wind tunnel will serve as a research tool on aerodynamics and will also allow other STEM students to explore their interests.

Design Brief

Client	Glenbrook South SELC, NASA.
Target Consumer	Ages 14+ who are interested in aerodynamics.
Designer	RKE Engineering LLC.
Problem Statement	Many people do not have access to a portable, affordable way to study aerodynamics, especially aerodynamic forces and airflow around objects.
Design Statement	Design and develop a tool that can be used to understand aerodynamics in an intuitive and engaging way. It should be constructed to be affordable and accessible, and its operation should not be extremely complicated.
Constraints	● Must be completed by May 1st, 2025 ● Total cost must be under $125 ● Must involve physics principles ● Must apply engineering principles

Research

Our first research focus was on the basic design behind wind tunnels as well as the physics principles behind them. We first looked at the different types of wind tunnels, choosing to focus on a subsonic open-return wind tunnel as it was the simplest and most common type used in scale testing. This type of wind tunnel is in essence, a tube with a fan at the end, pulling in air from the room and releasing that air back into the environment. The front of the wind tunnel consists of a large, open section, usually 6x the area of the test section, that gradually tapers down into the dimensions of the test area. This is called the contraction cone, and it is used to increase the velocity of the air. Next, the test section is where the model is placed for testing. This is the narrowest part of the tunnel and will include testing instruments. Finally, the last section is the diffuser section. This section tapers out from the narrow test section into a wider opening. This section includes the fan. The fan is placed at the end, "sucking" air into the tunnel instead of being placed at the front in order to reduce turbulence.

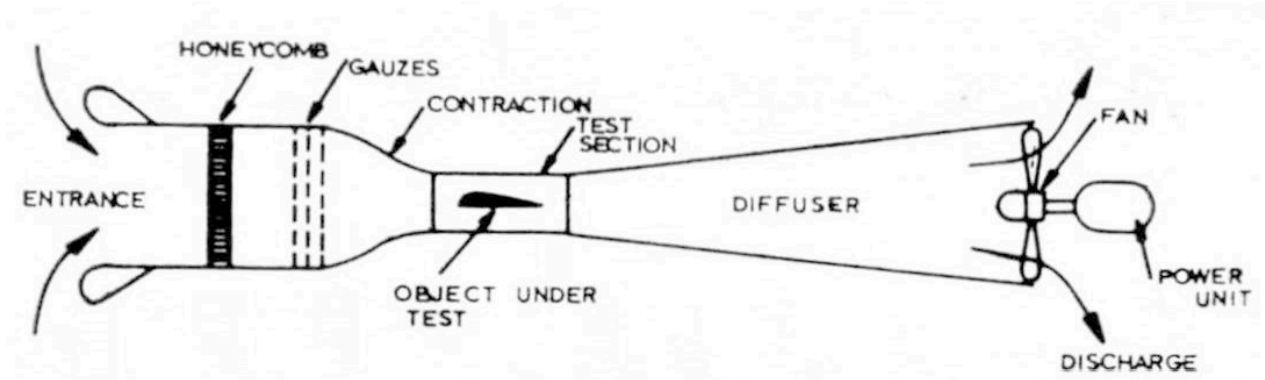
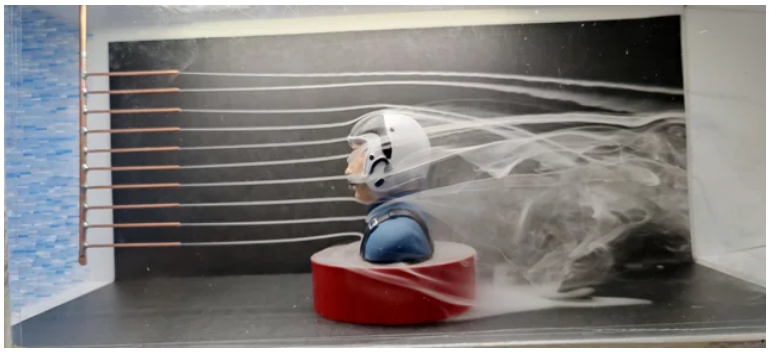


Fig. 1. Open circuit wind tunnel.

We then researched the different components required inside of a wind tunnel that optimizes performance and data collecting. One of the most important parts inside of the

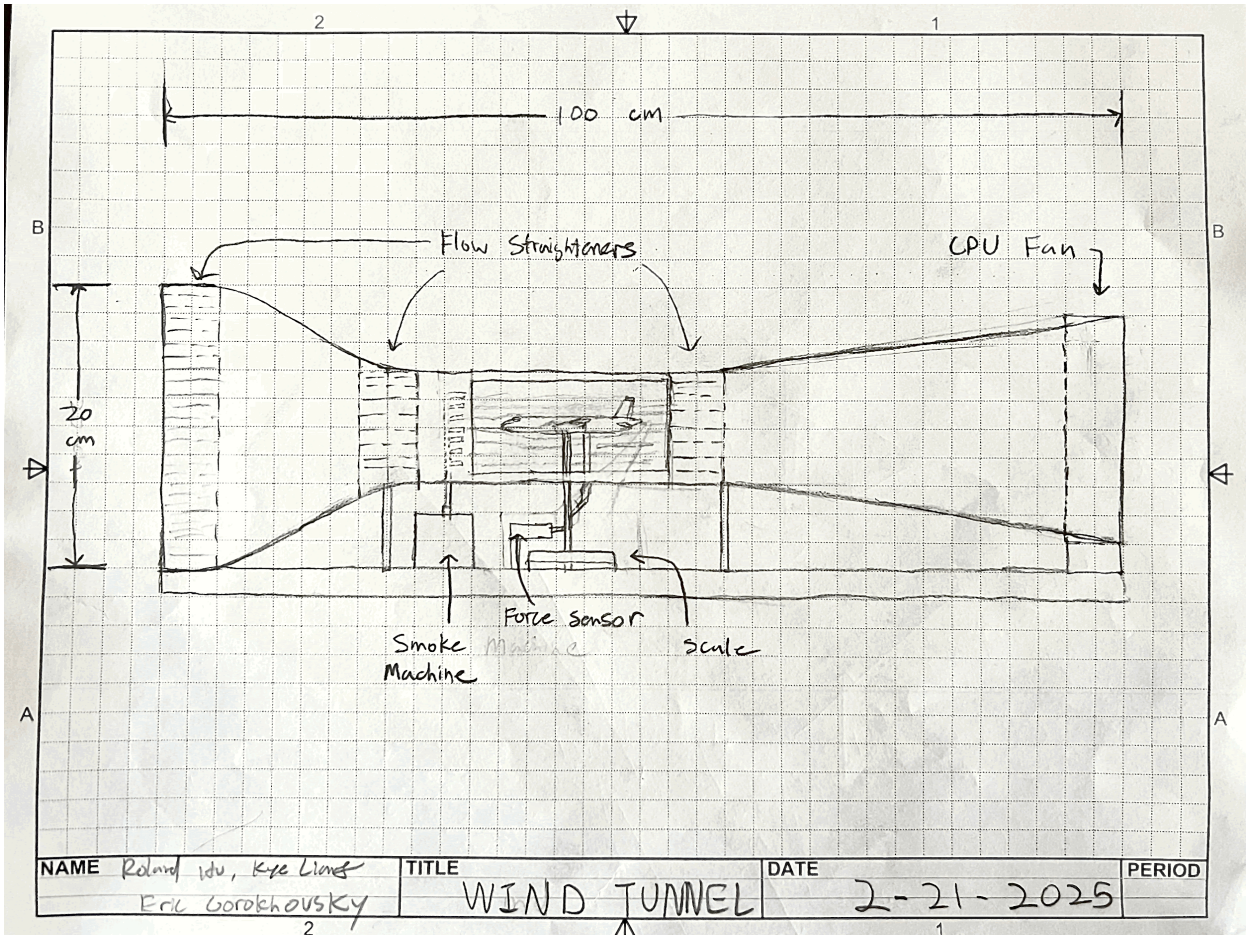
wind tunnel are the flow straighteners. When air enters, the air will not travel in straight, uniform lines throughout the tunnel. The air will be turbulent, which is not desirable for testing. To make the air flow smoothly ("laminar"), thin tubes (honeycomb screens) and meshes are placed along the tunnel across the entire area. These tubes serve to only allow air to flow straight, reducing turbulence. Another major component found in many tunnels is a smoke machine. This generates visible smoke, which is usually released in parallel streaks to visualize the airflow around an object. Finally, the data collecting instruments such as force sensors and scales are placed below the test section to gather important data such as lift and drag.



The final aspect of our research focused on finding other small-scale wind tunnels and their construction. We researched many DIY projects online and found that the majority of people used a 3-D printer in order to create the shell. Depending on the size, we may have to print individual pieces and use them to construct the outer shell. Since we are fortunate to have many Prusa printers, 3-D printing will be one of our smallest worries. Another design we found utilized more homemade appliances in order to create a functional wind tunnel. The outcome didn't look pretty but still worked. The design used a standard box fan and plastic in order to create a shell, similar to a wind tunnel. Furthermore, another design utilized a handheld miniature smoke machine to create the smoke. The handheld miniature smoke machines are mostly expensive which will not leave us room for other things like filament.

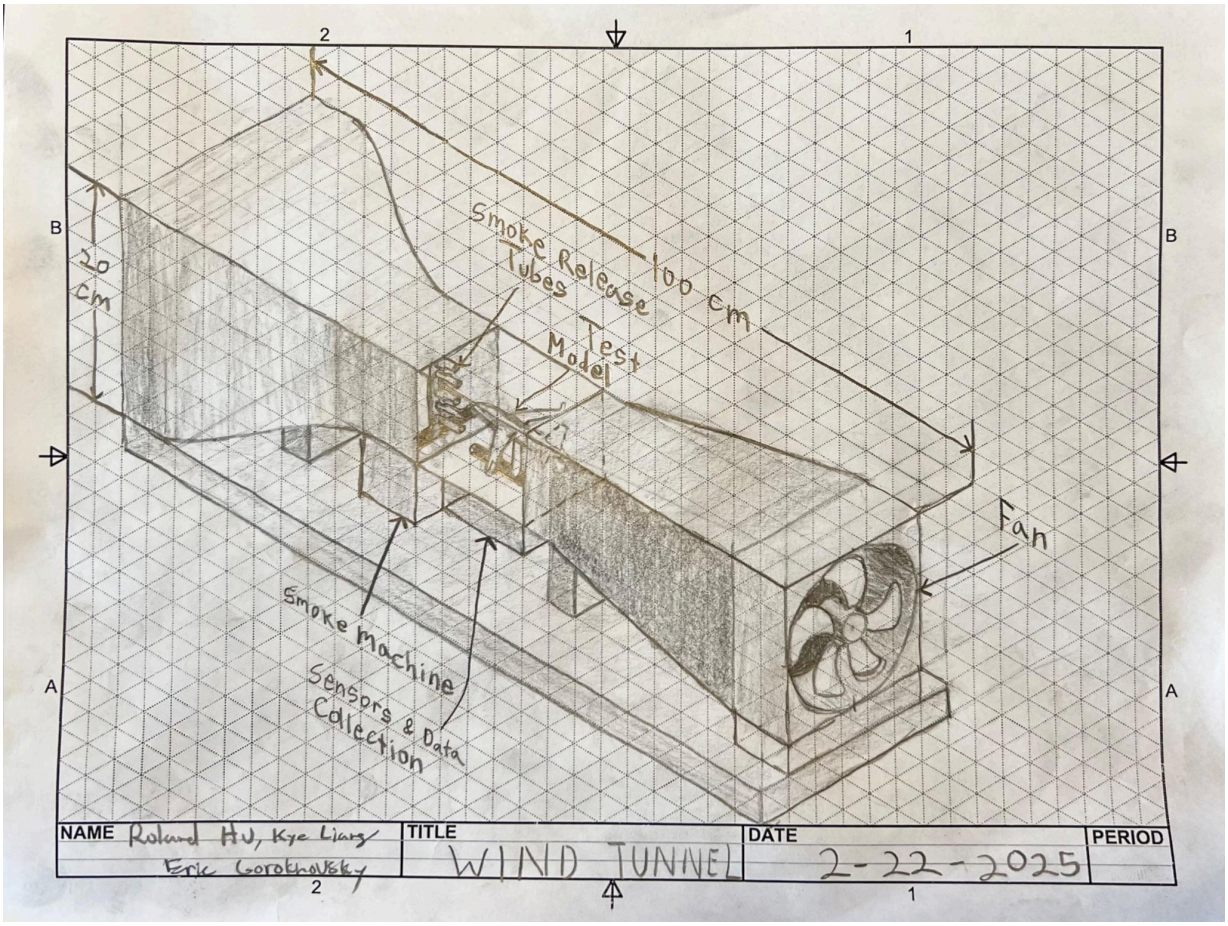
Overall, through our research, we learned many aspects of wind tunnels and their construction and components used, enabling us to have a solid understanding and plan of our own wind tunnel project.

Concept Sketches



Orthographic sketch of the wind tunnel

Roland Hu–02/21/2025



Isometric sketch of the wind tunnel

Roland Hu–02/22/2025

Physics Principles

The primary physics principle involved in the wind tunnel will be motion–in particular, fluid dynamics and forces.

We will make use of the continuity of flow, which states that the flow rate along any section of the wind tunnel must be constant since there is only one inlet and one outlet, to calculate the airspeed through the test section. In our designed testing envelope, the velocity of air though our wind tunnel is low enough ($v \ll 100\ m/s$) that air is essentially incompressible. This means the continuity equation $Av = Q = constant$ applies, where A is the cross sectional area, v is the airspeed, and Q is the volumetric flow rate. Given that we know the volumetric flow rate of the fan Q (technical specifications) and the cross sectional area A of any section, the airspeed though any section can be calculated as $v = \frac{Q}{A}$ (assuming turbulence is negligible).

The tunnel can then be used to deduce the aerodynamic properties of the test object such as the lift and drag of a plane or the downforce and drag of a car. These forces will be measured with a force sensor, and from this, properties such as the coefficient of lift C_l and the coefficient of drag C_d may be deduced for the test object. These coefficients encapsulate all the aerodynamic properties of the object (the geometry/shape, the texture, etc.) into a single number and as such, it is a very desirable metric to have. The equation $F = \frac{1}{2}CA\rho v^2$ describes the force of drag or lift on an object, where C is the coefficient of drag/lift, A is the cross-sectional area, ρ is the density of air and v is the airspeed. The coefficients can thus be calculated as $C = \frac{2F}{A\rho v^2}$.

Finally, the smoke will allow us to visualize airflow around an object, allowing many physical phenomena to be seen which is arguably the most interesting and largest demonstration of physics principles. For instance, we can visualize how air flows around airplane wings, and see many phenomena such as wingtip vortices or visualize the angle at which an airplane wing stalls. To give another example, vortex shedding which may induce unwanted vibrations in physical structures such as smokestacks can also be clearly seen. There are many physical phenomena that occur when air flows around objects, and through our wind tunnel, these can all be seen.

Overall, our wind tunnel incorporates and demonstrates many physics principles both qualitatively and quantitatively, allowing research into the aerodynamic properties of objects such as wings or cars and potentially aiding the development of new technologies as well as serving as a fun and engaging tool for students to explore new areas of interest.

Gantt Chart/Schedule

To organize our tasks and to keep track of our progress and deadlines, we used a Gantt chart. We listed the tasks to be completed and the approximate time frame of when each task will be done. Here is the Gantt chart for our project. This chart will provide strict deadlines on when to finish parts of the project.

Bill of Materials & Budget

We created a budget for our wind tunnel project so we could keep track of the money, parts, and where to purchase them. Our total cost is $153.05, and we are requesting $108.96 from the SELC. Here is our budget.

Tools and Equipment

Throughout the project, we will mostly use the 3D printers in order to create most of our parts. For example, we will 3-D print pieces of the wind tunnel and bolt them together to create our outer shell. If the filament option fails, we will then use wood which is more stable and reliable. To make certain cuts, we will use the table saw, miter saw, and wood glue/nails in order to connect the wood to create the shell. For the clear test section, we will use the laser cutter to cut sections of clear acrylic viewing windows. We will also use screwdrivers and screws to attach most of our parts like a fan to the inside of the shell. We might need to use a soldering iron in order to connect our project to an Arduino controller and breadboard (if needed).

Safety Concerns

Toxic smoke. The smoke used to visualize airflow could potentially cause health issues when inhaled and might irritate the lungs. To address these concerns, we will use non-toxic glycerin– or propylene glycol–based liquids in our smoke machine. In addition, the smoke output of our wind tunnel is relatively small compared to the theatrical/halloween smoke effects these machines are intended for, and as such, relatively little smoke will be released into the air and inhaled.

Rotating fan blades. We are planning to use an 8" duct fan to power our wind tunnel. At maximum speed, the blades will be rotating at 2500 rpm. If a person were to accidentally put a body part into the blades, they could potentially be injured. To resolve this issue, we will 3D print fan covers placed on both ends of the fan duct to prevent body parts from coming in contact with the blades.

Tool safety. While the majority of the parts will be 3D printed, some components may require cutting and/or drilling with the saws or the drill press. These tools can cause serious injury or death when used improperly. As such, we will carefully review the safety and operational guidelines of the tools we use, ensuring we are using the equipment correctly and safely.

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